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Top management team boundary-spanning behaviour, bricolage, and business model innovation

Shuai Yan^a, Baoliang Hu^a, Guang Liu^a, Xingjun Ru^a and Qitao Wu^b

^aSchool of Management, Hangzhou Dianzi University, Hangzhou, People's Republic of China; ^bSchool of Management, Qingdao University, Qingdao, People's Republic of China

ABSTRACT

Resources are basic element of business model innovation, but most enterprises face serious resource constraints due to their capability limitations. This study focuses on boundary-spanning behaviour of top management team and bricolage, to explore their influence mechanism on business model innovation. A hierarchical regression analysis is carried out on data from China. Responses from 163 enterprises indicate that top management team boundary-spanning behaviour has a significantly positive influence on business model innovation, that bricolage has a significantly positive effect on business model innovation, and that bricolage plays a significant intermediary role between top management team boundary-spanning behaviour and business model innovation. This study enriches theoretical and empirical researches on business model innovation while serving as a valuable reference for business practice.

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1. Introduction

Business model innovation (BMI) has attracted extensive attention from scholars and practitioners. BMI is not only the main means for enterprises to adapt to environmental changes and continuously create value (Karimi and Walter 2016), but also the key driving force for enterprises to create competitive advantages and achieve excellent performance (Martins, Rindova, and Greenbaum 2015). BMI enables incumbent enterprises to avoid the core competence trap and change competitive patterns, and also can help start-up enterprises to create new, untapped market space.

During BMI, enterprises often face resource constraints. Most new enterprises start with very limited resources (Ruef, Aldrich, and Carter 2003). Although some of these enterprises in high-growth industries are able to raise funds for BMI through venture capital and public capital markets, it's not widely applicable to most enterprises (Holtz-Eakin, Joulfaian, and Rosen 1994). Even for the established enterprises that have experienced growth, they often find it difficult to provide adequate resources for BMI (Baker and Nelson 2005). Therefore, how to break resource bottleneck for BMI has become a crucial practical problem.

The research on BMI employing the resource-based view (RBV) is an important branch of BMI theory, which emphasises the fundamental role of resources. For example, scholars who have discussed the elements of business model generally regard resources as a basic building block of business model (Johnson, Christensen, and Kagermann 2008; Osterwalder, Pigneur, and Tucci 2005), thus highlighting the important role of resources in BMI. However, there currently exist at least three gaps in BMI research based on RBV.

First, existing studies emphasise the importance of external resource acquisition to BMI. Most scholars focus on mergers and acquisitions (M&A), a typical way to obtain external resources, to study its impact on BMI (Ahuja and Katila 2001; Dezi et al. 2018). However, M&A often encounter many problems (Cartwright and Schoenberg 2006), such as high cost and risk; thus M&A are not universally feasible to overcome resource constraints for BMI. In contrast, boundary-spanning behaviour by the top management team (TMT) is a relatively low-cost and more flexible way to obtain external resources (Zhang and Ge 2016), so it is generally applicable to enterprises. TMTs are at the core of communication network between enterprises and the outside; thus it is relatively easy for them to conduct open and flexible boundary-spanning behaviour. Meanwhile, TMTs usually have high degree of network centrality and abundant external social capitals (Li, Lin, and Huang 2014), through which they can find some low-cost resources to alternate those high-cost by boundary-spanning behaviour. Thus TMT boundary-spanning behaviour is regarded as an important way for enterprises to break resource constraints by acquiring external resources (Zhang and Ge 2016). So, can TMT boundary-spanning behaviour positively affect BMI?

Second, existing studies also suggest that resource use is important to BMI. Most scholars have focused on the impact of resource integration on BMI (Peters 2016; Singaraju et al. 2016), as an important way of resource use. However, BMI through resource integration is premised on the fact that resources are adequate and accessible (Lii and Kuo 2016); it does not consider the scarcity of resources or difficulties in accessing them. Once resources are inadequate and inaccessible, BMI through resource integration becomes a challenge. Under resource constraints, bricolage is a more effective way to use resources (Baker and Nelson 2005), being regarded as a new resource use behaviour mechanism that copes with resources shortage. It stresses enterprises' subjective initiative to make creative use of existing resources at hand (Senyard et al. 2014). Therefore, compared with resource integration, bricolage places greater emphasis on creating something from nothing. So, can bricolage positively affect BMI?

Third, although previous studies have suggested that both resource acquisition and resource use can affect BMI, most of them separately explore their direct impact on BMI (Guo, Su, and Ahlstrom 2016); thus, they ignore the deep links between the two in BMI. This not only fails to reveal the process path of promoting BMI, but also leads enterprises not to effectively exert combined efforts of resource acquisition and resource use to solve resource constraints for BMI. The RBV theory emphasises that resource use depends on resource acquisition (Pfeffer and Salancik 2003). Therefore, this study attempts to establish the correlation between TMT boundary-spanning behaviour and bricolage, to explore whether boundary-spanning behaviour can affect BMI through bricolage.

Based on the above analysis, this paper aims to achieve three research objectives. First, it seeks to find a more feasible way of resource acquisition for enterprises to break resource constraints in BMI by examining the positive impact of TMT boundary-spanning behaviour on BMI. Second, it seeks to find a more effective way of resource use for enterprises to break resource constraints for BMI by examining the positive impact of bricolage on BMI. Third, it seeks to reveal the process path to promote BMI under resource constraints by examining the mediating effect of bricolage between TMT boundary-spanning behaviour and BMI.

2. Theoretical basis and research hypotheses

2.1. BMI based on the RBV

Resources play an important role in BMI. Casadesus Masanell and Zhu (2013) argue that BMI involves an innovation not only of products and procedures but also of the business model structure itself. Zott and Amit (2010) emphasise BMI as a process in which enterprises design new or improve existing operating systems by reorganising their resources and partners. Through resource acquisition and use, enterprises can achieve innovation in value creation, value proposition, and value acquisition (Claus 2017).

In the resource acquisition research, most studies focus on M&A to explore its effect on BMI. For example, Ahuja and Katila (2001) argue M&A can promote BMI by improving the absolute scale of knowledge base. But, M&A has its limitations, such as high cost and risk. In the resource use research, most studies focus on resource integration to explore its effect on BMI. For example, Salunke, Weerawardena, and McColl-Kennedy (2019) find that resource integration contributes to the development of innovation project. But resource integration ignores the resources scarcity and access difficulties.

2.2. TMT boundary-spanning behaviour and BMI

The concept of 'team boundary-spanning behaviour' was developed from the research on team boundaries (Marks, Zaccaro, and Mathieu 2000). Ancona and Caldwell (1992) found that team boundary-spanning behaviour has three dimensions: ambassador activities, task coordinator activities, and scout activities. This division has frequently been used by subsequent researchers. Therefore, based on the study of Ancona and Caldwell (1992), TMT boundary-spanning behaviour can be understood having the three dimensions: TMT ambassador activities, TMT task coordinator activities, and TMT scout activities (Zhang and Ge 2016). Through ambassador activities, TMT reports to higher departments or institutions to gain recognition and support. Through task coordinator activities, TMT solves development, operation, or design problems in collaboration with external organisations or teams. Through scout activities, TMT monitors similar projects by competitors, explores market opportunities and expertise information in external environment, and collects technical information outside.

This study argues that TMT boundary-spanning behaviour has three major impacts on BMI. First, The TMT is at the highest level of organisational structure and also the final decision maker and executor of BMI. So TMT ambassadorial behaviour may help to gain resource support for BMI from the director board, government departments, investors, shareholders, and relevant authorities (Sofka and Grimpe 2010; Sun and Zhou 2017). Second, one of the important functions of TMT is to coordinate various tasks and control decision making for business goals. The success of BMI often requires cooperation, collaboration, and information feedback from suppliers, purchasers, and other stakeholders. Thus, TMT task coordinator activities may help to drive BMI process through close and effective lateral communication (Meng, Wang, and Xu 2016). Third, compared with the other organisational members, TMT members tend to have higher developmental vision, stronger sense of innovation, and more sensitive market perception. Therefore, TMT scout activities could help to accurately capture market opportunities and valuable environmental information that, in turn, helps to improve the success probability of BMI (Geiger and Makri 2006). Thus, we hypothesise the following:

Hypothesis 1: TMT boundary-spanning behaviour has positive effects on BMI.

2.3. Bricolage and BMI

In the early years of its implementation, bricolage was often regarded as the last resort for enterprises dealing with imbalance in organisational planning and lack of competitiveness. However, with the deepening of enterprise cognition and change in competitive environment, bricolage has become a resource use strategy actively adopted by enterprises; further, it is regarded as an innovation strategy by SMEs in highly competitive and turbulent environment. For example, Fuglsang (2010) found that bricolage is a process strategy that inevitably leads to effective innovation.

Bricolage has been defined as making do by applying combinations of the resources at hand to new problems and opportunities (Baker and Nelson 2005, 7). The definition of bricolage contains three elements: existing resources at hand, making do, and combination of resources for new purposes. Existing resources include resources existing in enterprises or existing markets, but either not excavated or neglected; resources acquired by enterprises at a low cost through social exchange or non-contractual forms; and some unique strategic thinking and knowledge capabilities

possessed by individuals. It is based precisely on the extraordinary understanding of the use value of existing resources that bricolage can achieve to create something from nothing. Making do is a behavioural decision-making way to use existing resources to cope with new challenges or opportunities under resource constraints; it emphasises the improvisation of positive action. Further, it implies a bias toward action and active engagement with problems or opportunities rather than lingering over questions of whether a workable outcome can be created from what is at hand. Combining resources for new purposes involves the integration of resources with different strategic intentions and usage modes to meet new goals. For bricolage, Baker and Nelson(2005) identified five bricolage domains in which something can be created from nothing: physical inputs, labour, skills, customers, and the institutional environment.

In this study, we argue that bricolage has three major impacts on BMI. First, enterprises can create a new value proposition through bricolage by integrating existing and new resources. These resources may not be standardised, but they can be used to create new value proposition through bricolage. For example, Garud and Karnøe(2003) made a comparative study of wind turbine product innovation in Denmark and US. They found that Danish enterprises, even sought to obtain materials by searching waste products. Based on the materials available, they reinvented product designs and achieved the expected goals. The enterprises in US, by contrast, focused on how to make and improve components and parts based on traditional designs, and ended up doing less well than their Danish counterparts. This case provides a good reference source for enterprises that use resources step by step and indicates that those seemingly useless resources can also play an unexpected role in achieving innovative products or services through bricolage. Therefore, bricolage can drive BMI through innovating value proposition of business model.

Second, the new value proposition created by bricolage may lead enterprises to go beyond their market boundaries, and re-analyse the target market segment and reposition themselves. In this process, new market segments, value delivery channels, and service modes may be formed (Desa and Basu 2013), which are all manifestations of BMI. In addition, by providing products or services that would otherwise be unavailable to customers (because of poverty, thriftiness, or lack of availability), enterprises can continuously create and expand market segments, and even become potential disrupters in the mainstream market (Baker and Nelson 2005). Thus, bricolage can promote BMI by changing 'customer segments', 'channels' and 'customer relationship' of business model.

Third, by refusing to execute the default industry regulations and traditional practices, bricolage may help enterprises to form new regulations and norms for the existing situation (Baker and Nelson 2005). The new regulations and norms produced by bricolage often include a new organisational structure, new production process, new trading mode, new customer demand solution and so on. All these, in turn, will lead to BMI. In short, bricolage can promote BMI by changing one or more elements of business model. We thus hypothesise the following:

Hypothesis 2: Bricolage has positive effects on BMI.

2.4. TMT boundary-spanning behaviour and bricolage

According to the statistical analysis method, the precondition to test the mediating effect of bricolage is to test whether the boundary-spanning behaviour can positively affect bricolage. So the relationship between TMT boundary-spanning behaviour and bricolage should be discussed.

In bricolage, existing resources include not only the resources that enterprises own, but also those obtained from social network partners in easy ways and at low costs. TMT boundary-spanning behaviour is an important way for enterprises to form social networks. The stronger the TMT boundary-spanning behaviour, and the larger the scale of social networks owned by enterprises, the easier for them to establish links with more network members who own various kinds of technology, capital and knowledge resources. The more resource nodes enterprises own, the wider the range of resources that can be acquired and mobilised. Therefore, TMT boundary-spanning behaviour

can help to find more alternative and utilisable resources for bricolage. For example, Baker(2007) did a case study of a toy store in US, and found that the store could use local resources as cheaply as possible by tapping into all the social relationships to which it had access.

Moreover, TMT boundary-spanning behaviour can help enterprises to remove the 'lock-in' restrictions on the use of existing resources and discover other potential uses of the resources at hand (Di Domenico, Haugh, and Tracey 2010), through information interaction, task coordination and environmental detection. Therefore, TMT boundary-spanning behaviour can contribute to the redefinition of the uses and attributes of resources, thereby promoting more bricolage activities. We thus hypothesise as follows:

Hypothesis 3: TMT boundary-spanning behaviour has positive effects on bricolage.

2.5. Mediation effect of bricolage

TMT boundary-spanning behaviour, through transaction or exchange, can satisfy a certain degree of resource demand for BMI. However, enterprises are inevitably constrained by certain subjective initiatives in the use of resources; this makes it difficult for them to make BMI. Thus, TMT boundary-spanning behaviour needs to stimulate the creative use of resources through bricolage to promote BMI. In addition, TMT boundary-spanning behaviour of many enterprises serves only to build a resource network without directly leading to innovations in their business models. So, enterprises must re-identify, redefine, recombine, and creatively use various resources in their resource network. We thus hypothesise the following:

Hypothesis 4: Bricolage mediates the relationship between TMT boundary-spanning behaviour and BMI.

Figure 1 displays the conceptual model to be examined in the current study.

3. Research methodology

3.1. Sample and data collection

Referring to previous literatures, we designed the questionnaire. Because respondents were from China and the questionnaire items were all from English-language literatures, we firstly translated items into Chinese-language and then back-translated into English-language to avoid language misunderstanding. Subsequently, we consulted two scholars in BMI and three top managers from enterprises to revise the questionnaire. Finally, five top managers were selected for a small-scale test of questionnaire. According to their responses, we further modified and refined the questionnaire.

The sample frame for this study consisted of the small and medium-sized enterprises (SMEs). In China, SMEs account for 97% of the total. The division standard of SMEs is from the China official document 'Small and medium-sized enterprises division standard (2011)'. They mainly concentrate in the five economically developed regions of Shanghai, Beijing, Guangdong, Zhejiang and Jiangsu. Meanwhile, they are the representative regions in China with relatively high comprehensive

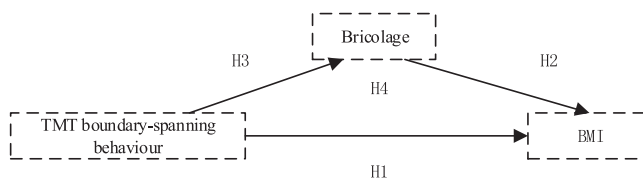


Fig.1. Conceptual Model

Figure 1. Conceptual model.

innovation capability. So most enterprises in these regions show strong BMI vitality for development and competitive advantages. Therefore, the empirical investigation was conducted in these regions.

The survey was conducted from July 2017 to June 2018. Based on the SMEs directory of Shanghai, Beijing, Guangdong, Zhejiang and Jiangsu, 300 enterprises were selected by simple random sampling. In order to maximise the response rate, we contacted at least one top manager from each sample enterprise by phone before conducting the formal survey. We used face-to-face interviews, emails, phone calls and social software to collect data. Finally, 192 enterprises were interviewed in which 29 were excluded due to inadequate completion of survey instruments, for an effective response rate of 54.3% (163/300). A descriptive statistical analysis of the effective samples is shown in Table 1. As shown in Table 1, the sample enterprises are diverse in age, size and industry and evenly distributed, which is consistent with the overall situation of SEMs in China. Therefore, although we adopt the simple random sampling method, the enterprise sample is sure to be representative.

To test non-response bias, we used extrapolation method to examine the differences between the responding and non-responding enterprises (Armstrong and Overton 1977). According to the time sequence, two groups of the questionnaire were selected for *T*-test. The first group was 30 copies recovered earlier and the second group was 30 copies recovered latest. The test results showed that there was no significant differences in age, size and region between the respondents and the non-respondents ($p > 0.05$).

The common method bias (CMB) was another potential threat to impact research results, caused by the same measurement environment, item context, and item content features. Thus, Harman's one-factor test was implemented to examine CMB. The results showed that all the measurement items were divided into five factors with eigenvalues higher than 1.0, which explained 70.011% of the total variance. The first factor explained only 32.848% of the variance, less than the benchmark value of 50.0% (Harman 1976). These results suggested that CMB was unlikely to be a severe threat in this study.

3.2. Measurement of variables

The original measurement scale of team boundary-spanning behaviour was designed by Ancona and Caldwell (1992), including three dimensions of ambassador activities, task coordinator activities, and scout activities. The original scale is widely used (Faraj and Yan 2009), so we draw nine items from it to measure TMT boundary-spanning behaviour. The TMT ambassador activities are measured by three

Table 1. Descriptive statistics of respondents.

	Frequency	%		Frequency	%
Region			Age (years)		
Shanghai	32	19.6	1–5	37	22.7
Beijing	24	14.7	6–10	31	19.0
Guangdong	28	17.2	11–15	22	13.5
Zhejiang	35	21.5	16–20	16	9.8
Jiangsu	44	27.0	21–25	14	8.6
Industry			26–30	9	5.5
light industrial machinery	22	13.5	≥31	34	20.9
biological medicine	14	8.6	Size (persons)		
wholesale and retail	10	6.1	1–20	17	10.4
warehousing and logistics	9	5.5	21–50	21	12.9
finance	11	6.7	51–100	15	9.2
fast moving consumer goods	12	7.4	101–200	19	11.7
electronic information technology	24	14.7	201–300	11	6.7
education/ scientific research/ training	20	12.3	301–500	9	5.5
e-commerce	18	11.0	1000 ≥ 501	71	43.6
advertising/creative	13	8.0			
media communication	10	6.1			

items (1–3). The TMT task coordinator activities are measured by three items (4–6). The TMT scout activities are measured by three items (7–9).

Senyard, Baker, and Davidsson(2009) developed the measurement scale of bricolage based on prior research and a multi-dimensional definition of bricolage by Baker and Nelson(2005), which includes existing resources at hand, making do, and combining resources for new purposes. This scale was developed following standard protocols for scale development and has been widely adopted (Gundry et al. 2011). So we used their measurement scale to measure bricolage, including eight items.

Few scales for BMI are available. Zott and Amit (2007) have developed a framework for measuring efficiency-centred and novelty centred business models. However, Pedersen, Gwozdz, and Hvass (2018) argued that BMI should be considered as a matter of degree rather than either–or issue. So they proposed a continuum between the exploiting of existing competences and the exploration of new business opportunities. Thus they developed a new measurement scale for BMI, including nine items. These items correspond to the nine components of business model canvas: value proposition, customer segments, key resources, key activities, key partnerships, customer relationships, channels, cost structure, and revenue streams. Therefore, we draw the nine measurement items to measure BMI.

3.3. Reliability and validity

Composite reliability assesses inter-item consistency, usually estimated with Cronbach's alpha. As can be seen from Table 2, the lowest alpha value is 0.927, exceeding the cut-off point of 0.700(Nunnally, Bernstein, and Berge 1967). Convergent validity is the extent to which the items on a scale measure

Table 2. Measures and validation.

Item	Loading
TMT boundary-spanning behavior (Alpha = 0.927; CR = 0 .944; AVE = 0 .655)	
1. Report the progress of team to a higher organisational level.	0.722
2. Persuade other individuals that the team's activities are important.	0.835
3. Acquire resources (e.g. money, new members, equipment) for team or enterprise.	0.811
4. Resolve development/operation/design problems with external groups.	0.768
5. Coordinate activities with external groups.	0.796
6. Review development/operation/product designs with outsiders.	0.821
7. Find out what competing firms or groups are doing on similar projects.	0.849
8. Scan the environment inside or outside the organisation for marketing ideas/expertise.	0.930
9. Collect technical information/ideas from individual outside the team.	0.730
Bricolage (Alpha = 0 .956; CR = 0 .963; AVE = 0.767)	
1. We are confident of our ability to find workable solutions to new challenges by using our existing resources.	0.903
2. We gladly take on a broader range of challenges than others with our resources would be able to.	0.868
3. We use any existing resource that seems useful to responding to a new problem or opportunity.	0.890
4. We deal with new challenges by applying a combination of our existing resources and other resources inexpensively available to us.	0.862
5. When dealing with new problems or opportunities we take action by assuming that we will find a workable solution.	0.841
6. By combining our existing resources, we take on a surprising variety of new challenges.	0.884
7. When we face new challenges we put together workable solutions from our existing resources.	0.913
8. We combine resources to accomplish new challenges that resources weren't originally intended to accomplish.	0.843
BMI (Alpha = 0.940; CR = 0.950; AVE = 0.677)	
1. Over the past three years, Focus is on developing radically NEW products and/or services.	0.873
2. Over the past three years, Focus is on identifying and serving entirely NEW markets and customer segments.	0.855
3. Over the past three years, Focus is on developing and/or acquiring NEW resources and competences.	0.881
4. Over the past three years, Focus is on developing NEW core processes and activities.	0.841
5. Over the past three years, Focus is on establishing relationships with NEW strategic business partners.	0.843
6. Over the past three years, Focus is on developing NEW tools for building customer relationships.	0.820
7. Over the past three years, Focus is on selling products and/or services through NEW channels.	0.817
8. Over the past three years, Focus is on making MAJOR changes in the combination of costs incurred when operating the company.	0.740
9. Over the past three years, We have developed NEW ways of generating revenue.	0.722

the abstract or theoretical construct, estimated with factor loading. Table 2 shows that all loadings are higher than the 0.700 benchmark (Nunnally, Bernstein, and Berge 1967). We also ran confirmatory factor analysis (CFA) to further test composite reliability and convergent validity. All composite reliability (CR) values, at 0.944 or higher, exceed the cut-off point of 0.700 (Fornell and Larcker 1981). The average variances extracted (AVE) for each construct are 0.655 or higher, exceeding the 0.500 benchmark (Fornell and Larcker 1981). Thus, the CFA results show that composite reliability and convergent validity are confirmed acceptable.

Discriminant validity refers to the degree to which the measures of a construct are unrelated between each others (Hatcher 1994). Table 3 shows that the square root values of AVE for each latent variable are all larger than the correlation between any pair of constructs, thus confirming discriminant validity acceptable (Wang et al. 2016).

4. Data analysis and results

Hierarchical regression analysis can reveal direct and mediating relationships between variables (Baron and Kenny 1986). This method has often been applied in business model studies (Hu 2014; Wei, Zhao, and Zhang 2014). Thus we conducted a hierarchical regression analysis for all the hypotheses. Four regression models should be estimated: first, regressing BMI on TMT boundary-spanning behaviour; second, regressing BMI on bricolage; third, regressing bricolage on TMT boundary-spanning behaviour; forth, regressing BMI on both TMT boundary-spanning behaviour and bricolage.

Table 4 presents the main hierarchical regression results. Model 1 examines the impact of TMT boundary-spanning behaviour on BMI (hypothesis 1). The F -value is 24.351 ($p < 0.001$), indicating that Model 1 has statistical significance. The highest variance inflation factor (VIF) value in Model 1 is 1.778, which is lower than the critical value of 10 (Jansen et al. 2008), indicating that Model 1 does not exhibit multicollinearity. The results show that TMT boundary-spanning behaviour $\beta = 0.616^{***}$, $p < 0.001$ has significantly positive effects on BMI. Thus, hypothesis 1 is supported.

Model 2 examines the impact of bricolage on BMI (hypothesis 2). The F value is 26.095 ($p < 0.001$), indicating that model 2 has statistical significance. The highest VIF value in Model 2 is 1.749, which is lower than the critical value of 10, indicating that Model 2 does not exhibit multicollinearity. The results show that bricolage $\beta = 0.630^{***}$, $p < 0.001$ has significantly positive effects on BMI. Thus, hypothesis 2 is supported.

Model 3 examines the impact of TMT boundary-spanning behaviour on bricolage (hypothesis 3). The F -value is 45.510 ($p < 0.001$), indicating that Model 3 has statistical significance. The highest VIF value in Model 3 is 1.778, which is lower than the critical value of 10, indicating that Model 3 does not exhibit multicollinearity. This results show that TMT boundary-spanning behaviour $\beta = 0.652^{***}$, $p < 0.001$ has significantly positive effects on bricolage. Thus, hypothesis 3 is supported.

Model 4 examines bricolage's mediating effect between TMT boundary-spanning behaviour and BMI (hypothesis 4). The F -value is 25.582 ($p < 0.001$), indicating that Model 4 has statistical significance. The highest VIF value in Model 4 is 2.160, which is lower than the critical value of 10, indicating that Model 4 does not exhibit multicollinearity. This results show that, when TMT boundary-spanning behaviour and bricolage are introduced together, the impact of bricolage on BMI is still significant (β

Table 3. Descriptive statistics and correlations.

Construct	Means	SD	1	2	3	4	5
1.Firm age	3.61	2.192					
2.Firm size	4.85	2.234	0.639**				
3.Boundary-spanning behavior	5.58	0.935	0.083	0.224**	0.809		
4.Bricolage	5.51	0.934	0.184*	0.258**	0.471**	0.876	
5.BMI	5.21	1.016	0.139	0.161*	0.457**	0.510**	0.823

Note: + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Means are measured based on average factor scores; SD means standard deviation; Values in the diagonal row (bold) are the square roots of AVEs and the others are the correlations between constructs.

Table 4. Regression results.

	Model 1	Model 2	Model 3	Model 4
Firm age	0.125(1.529)	0.040(.496)	0.107(1.513)	0.083(1.071)
Firm size	0.056(0.673)	0.027(0.027)	0.032(0.442)	0.069(0.869)
Region	0.061(0.967)	0.021(0.021)	0.099(1.804)	0.023(0.376)
TMT boundary-spanning behaviour	0.616*** (9.498)		0.652*** (12.617)	0.341*** (3.922)
Bricolage		0.630*** (9.848)		0.389*** (4.482)
F	24.351***	26.095***	45.510***	25.582***
R ²	0.381	0.398	0.535	0.452
Adjusted R ²	0.366	0.383	0.524	0.434
VIF	≤1.778	≤1.749	≤1.778	≤2.160

⁺ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

= 0.389***, $p < 0.001$), while the impact of TMT boundary-spanning behaviour on BMI is weakened significantly ($\beta = 0.341$ ***, $p < 0.001$), demonstrating that bricolage mediates the relationship between TMT boundary-spanning behaviour and BMI. Thus, hypothesis 4 is supported.

5. Discussion and implications

5.1. Summary of conclusions

This study discusses the influence mechanism of TMT boundary-spanning behaviour and bricolage on BMI. It draws three conclusions as follows.

First, through testing hypothesis 1, it concludes that TMT boundary-spanning behaviour has a significant positive impact on BMI. This conclusion further confirms the important role of TMTs for BMI (Svejenova, Planellas, and Vives 2010). It highlights that TMT boundary-spanning behaviour is a key antecedent factor of BMI.

Second, through testing hypothesis 2, it finds that bricolage has a significant positive impact on BMI. This conclusion is similar with other findings, such as the bricolage's impact on product innovation (Wu, Liu, and Zhang 2017), on service innovation (Salunke, Weerawardena, and McColl-Kennedy 2013). It shows that bricolage is another important antecedent factor of BMI.

Third, through testing hypothesis 3 and 4, it reveals that TMT boundary-spanning behaviour has a significantly positive impact on bricolage and that bricolage plays mediating effect between TMT boundary-spanning behaviour and BMI. This conclusion is similar to that of Baker and Nelson (2005) and Di Domenico, Haugh, and Tracey (2010); however, they did not conduct in-depth research on the mediating effect of bricolage. This study shows that there is a deep relation between TMT boundary-spanning behaviour and bricolage in BMI.

5.2. Theoretical contributions

This study makes several important theoretical contributions. First, conclusion 1 enriches and deepens the theory on relationship between external resource acquisition and BMI. M&A and TMT boundary-spanning behaviour are important ways of external resource acquisition, but TMT boundary-spanning behaviour is lower cost, more flexible and more universal than M&A under resource constraints. Previous scholars have studied the positive influence of M&A on BMI (Ahuja and Katila 2001; Dezi et al. 2018), and some scholars explored the influence of TMT boundary-spanning behaviour on enterprises innovation performances (Zhang and Ge 2016). However, the effect of TMT boundary-spanning behaviour on BMI is ignored. Therefore, the study shows the positive influence of TMT boundary-spanning behaviour on BMI; thus, it provides a more feasible choice for enterprises to solve resource acquisition problems for BMI.

Second, conclusion 2 enriches and deepens the theory on relationship between resource use and BMI. Both resource integration and bricolage are important ways of resource use; but bricolage is

more effective than resource integration under resource constraints, because bricolage emphasises the creative use of existing resources. Previous studies have studied the positive influence of resource integration on BMI (Lii and Kuo 2016). However, its main premise is resource adequacy and accessibility. By ignoring resources scarcity and access difficulties, resource integration faces a challenge when dealing with resource constraints. Therefore, the study shows the positive impact of bricolage on BMI; it powerfully explains why different enterprises could create heterogeneous business models by using the same limited resources.

Third, conclusion 3 finds the relationship between TMT boundary-spanning behaviour and bricolage and its effect on BMI; it reveals the path to promote BMI. Previous studies have emphasised the importance of resource acquisition and resource use for BMI; however, most only discussed the direct impact of each resource technique on BMI (Guo, Su, and Ahlstrom 2016), and ignored their correlation and its effect on BMI. The study shows the positive impact of TMT boundary-spanning behaviour on bricolage and further shows the intermediating effect of bricolage between TMT boundary-spanning behaviour and BMI, thereby laying the theoretical foundation to open the black box of BMI process.

5.3. Managerial implications

This study has significant practical implications for enterprises. First, enterprises should give full play to the work potential of TMTs as the core groups with strategic decision-making responsibility, and encourage them to carry out more extensive boundary-spanning behaviour for creating valuable resource network. A targeted resource acquisition mechanism should also be established to effectively identify and manage boundary resource networks for acquiring resource support at low costs and exploring BMI opportunities.

Second, enterprises should identify resource use value beyond traditional attributes and make resource use decision in a more flexible and active manner. They should improve the sensitivity and familiarity of resource inputs and to create more new value by bricolage. They should also be encouraged to make more customer bricolage and more institution bricolage. Moreover, it is necessary for enterprises to take necessarily proactive steps to overcome more barriers, such as hesitation, doubts, and prejudices, to resource use.

Third, it is necessary for enterprises to establish some effective mechanisms to realise the integration of TMT boundary-spanning behaviour and bricolage. For example, enterprises can establish some procedural mechanisms such as information sharing, discussions and communication to identify the potential resource use after TMT boundary-spanning behaviour. Moreover, enterprises may design some effective trial-and-error learning mechanisms for resource use to cooperate with resource network exploitation.

5.3. Limitations and future research

This study has several limitations that can serve as opportunities for further research. First, it adopts cross-sectional data, without considering the change in enterprise behaviour over time. Therefore, we will conduct longitudinal case studies based on some e-commerce platform enterprises and manufacturing enterprises in China to supplement the present study. Second, it only explores the relevant research hypotheses within one country, without considering the influence of cultural differences. Therefore, we will consider collecting more enterprise samples from different countries such as America and China, to conduct a transnational comparative study. Third, it does not take into account the influence of strategic orientation on bricolage. Because enterprises with different strategic orientations focus on different resource types, there will be differences in their bricolage strategies. Thus, we will introduce strategic orientation to test its moderating effect with the enterprise sample in Yangtze river delta region of China, thereby further exploring the BMI process.

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Notes on contributors

Shuai Yan is a lecturer of Innovation & Entrepreneurship at School of Management, Hangzhou Dianzi University, China. She received her PhD in Management from Hohai University, China. She was a Visiting Scholar at Kentucky University in 2012. Her current research interests mainly focus on business models and technological innovation.

Baoliang Hu (corresponding author) is a professor of Innovation Management at School of Management, Hangzhou Dianzi University, China. He received his PhD in Management from Zhejiang University, China. He was a Visiting Scholar at Worcester Polytechnic Institute in 2012. His research work has been published in European Management Journal. His current research interests mainly focus on business models and technological innovation.

Guang Liu is an associate professor of Innovation Management at School of Management, Hangzhou Dianzi University, China. He received his PhD in Management Science and Engineering from Donghua University, China. His current research interests mainly focus on knowledge management and business model innovation.

Xingjun Ru is a lecturer of Green Innovation at School of Management, Hangzhou Dianzi University, China. She received her PhD in Management Science and Engineering from Chinese Academy of Sciences, China. Her current research interests mainly focus on green innovation.

Qitao Wu is a lecturer of Supply Chain Management at International Business School, Qingdao University, China. He received his PhD in Environmental Engineering from Ocean University of China. His current research interests mainly focus on green innovation.

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